

TY CSE AY-2026-27 Sem-I

Sub: Cutting Edge Technologies Lab (7CS352)

Assignment No 2 (Operators and Control flow)

Due date-31/07/2026

1. Student Grade Management System

Write a Swift program to calculate a student's total marks and percentage for three subjects. Use arithmetic and assignment operators to calculate the result. Use comparison and logical operators with if-else statements to determine whether the student has passed or failed. Assign a grade using if-else-if or switch.

2. Electricity Bill Calculator

Write a Swift program to calculate an electricity bill based on the number of units consumed. Use arithmetic and assignment operators to calculate the bill amount. Use if-else statements to apply different rates for different unit ranges and display the final bill.

3. Simple Calculator

Write a Swift program that accepts two numbers and an arithmetic operator (+, -, *, /). Use a switch statement to perform the selected operation and display the result. Handle division by zero using a control flow statement.

4. ATM Withdrawal System

Write a Swift program that simulates an ATM withdrawal. The program should check whether the entered PIN is correct and whether the requested withdrawal amount is within the available balance. Use comparison and logical operators along with if-else statements to approve or reject the transaction.

5. Shopping Discount Calculator

Write a Swift program to calculate the final price of products in a shopping cart. Use arithmetic operators to calculate the total amount and apply discounts based on the purchase value using if-else-if statements. Display the original amount, discount, and final payable amount.

6. Employee Salary Calculator

Write a Swift program to calculate an employee's final salary based on basic salary, allowances, and deductions. Use arithmetic and assignment operators for calculations. Use if-else statements to determine a bonus based on the employee's performance rating.

7. Movie Ticket Booking System

Write a Swift program to calculate the total ticket price based on the number of tickets and ticket type. Use a switch statement to determine ticket prices for regular, premium, and VIP categories. Apply a discount if the customer purchases more than a specified number of tickets.

8. Parking Fee Calculator

Write a Swift program to calculate parking charges based on the number of hours a vehicle is parked. Use if-else-if statements to apply different charges for different durations. Use arithmetic and assignment operators to calculate the total fee.

9. Number Classification Program

Write a Swift program that accepts an integer and determines whether it is positive, negative, or zero. Also check whether the number is even or odd using comparison and arithmetic operators with appropriate control flow statements.

10. Student Attendance Eligibility System

Write a Swift program to calculate a student's attendance percentage. Use comparison and logical operators to determine whether the student is eligible to attend the examination. The student should be eligible if their attendance is above the required percentage or if they have an approved medical exemption.